

NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. Centered continuation errors connect approximation, changes in the economic environment, and full-policy loss. A constructive nonlinear implementation trains each continuation against its own fitted future and returns either an all-state policy certificate or a recorded failure. A retained-policy theorem separates inadequate verification from inadequate policies: seven previously unresolved targets are certified without changing any network or deployed action. A paired residual identity permits direct policy-cost comparisons under continuous innovations, with simultaneous finite-sample and numerical error control. The complete training catalogue retains strong spline and ridge comparators and all adverse findings. The results identify the distinction between producing a policy, certifying it, and establishing its comparative economic or computational value. The controlled-diffusion, recursive-preference, and game formulations and their application-specific conditions remain part of the paper.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Recursive utility, Computational economics.

1. INTRODUCTION

An economic decision uses a continuation value because its consequences extend beyond the current period. A learned continuation is useful only to the extent that the decisions made with it are accurate. Prediction loss, a Bellman residual at sampled states, and the value of the policy actually implemented are different objects. Their distinction is particularly important when a counterfactual changes future policies or when a recursive preference evaluates the continuation generated by those policies rather than the optimum.

This paper develops Neural Bellman Operators (NBOs) as a policy-evaluation and feasible-improvement method. An operator takes economic primitives, a current policy, and computational resources as inputs. It returns a continuation, a feasible policy, and an economic error account, or it records that the requested account was not obtained within its cap. The continuation is evaluated against its own fitted future. Current actions, policy-specific future values, numerical execution, and the cost of verification enter the same account. The construction is not a proposal to rename an arbitrary neural predictor as a Bellman operator.

There are three parts to the argument. First, only action-dependent continuation errors affect a current decision. Their oscillation, rather than their absolute level, controls loss. A transport bound identifies when an anchor continuation remains valid after the future economy changes. These results connect reuse and refresh decisions to economic tolerances. They also preserve the original formulation in controlled diffusions, recursive preferences, endogenous preferences, and games, subject to the distinct hypotheses of those applications.

Second, a signed residual certificate controls a complete returned policy. Its upper bound contains the widths of the residual enclosures, the suboptimality of the selected actions, and the terminal enclosure. It is invariant to additive date-specific shifts of the fitted continuation. This leads to a useful separation: a policy can be accurate even though a coarse verifier does not establish its accuracy. We prove that residual verification can be refined while retaining the original network, the original deployed actor, and its original allowance. The resulting bound measures improved knowledge of the same policy, not a policy improvement disguised as additional verification.

Third, certification of each policy relative to the optimum does not rank two policies. We construct their cost difference directly. Under expected-cost preferences, an own-continuation Bellman telescoping identity produces an unbiased policy-cost score with a support width governed by the signed certificate. Coupling innovations across policies and enclosing every simulated path gives simultaneous finite-sample intervals for the difference in the original continuous-innovation economy. The statistical assertion is conditional on frozen policies and the stated independent

simulation model; outward arithmetic and unresolved action-cell boundaries have their own explicit treatment.

The numerical implementation starts from economic primitives. In the scalar investment economy, it fits a fresh ReLU continuation backward against targets using its own previously fitted future network. In the two-capital economy, it trains hidden locations in a convex squared-ReLU continuation, with continuous simplex actions and continuous uniform innovations. The second experiment includes a flexible nonnegative ridge dictionary as well as a quadratic projection. Each comparator constructs its own future targets and uses the same final economic criterion. Neither conventional values nor inherited neural weights are used as labels for the neural services.

We report the complete 150-service catalogue, not only successful candidates. In the original capped scalar services the neural procedure reaches the all-state target 0.06 in all twelve cases and 0.04 in six, whereas the spline reaches both targets in every case. In the coupled services the neural procedure reaches 0.25 in six cases and 0.10 in five; the flexible ridge reaches both in all six. These statements describe the original service, including its original verification cap.

A subsequent diagnostic holds both the neural weights and the deployed policies fixed. Refining only residual verification certifies all six previously unresolved scalar policies at 0.04 and the previously unresolved coupled policy at 0.10. The original failures are not relabeled. The diagnostic shows that they do not, by themselves, identify a failure to construct an accurate neural policy. It does not identify the population success probability of the optimizer, establish a new end-to-end runtime, or change any policy's cost.

The comparative findings are equally integral to the paper. The original scalar signed policy comparisons frequently favor the spline. The new coupled direct intervals all contain zero and therefore do not order the policies. The precision controller genuinely escalates after rejected low-precision proposals, but its complete recorded work does not improve on fixed binary64 in this experiment. These findings distinguish the certified capability of the construction from a comparative advantage that would require additional evidence.

The paper retains its original economic subject and full theory. The exposition now follows the economic problem, the operator, the policy theorem, and the evidence. The complete previous article, proof supplement, and economic applications remain unchanged and linked as companions; the preservation map records their exact identities. No historical theorem or application is deleted to obtain a shorter active exposition. The present article and new technical supplement state and prove the additional results used by the new evidence.

1.1. *Relation to the literature*

Neural Bellman Operators sit within approximate dynamic programming, fitted-value methods, and policy iteration, not outside them. Regression-based continuation methods and simulation-based dynamic programming supply important conventional benchmarks (Judd, 1998, Powell, 2011, Longstaff and Schwartz, 2001, Ernst et al., 2005). Actor–critic and approximate policy-iteration methods separate evaluation and improvement (Konda and Tsitsiklis, 1999, Scherrer et al., 2015); safe-improvement arguments likewise distinguish a proposed action from a verified gain (Kakade and Langford, 2002, Pirodda et al., 2013). We use these distinctions to organize the economic object returned by the algorithm.

Neural methods for differential equations provide ways to approximate values and their derivatives (Sirignano and Spiliopoulos, 2018, Han et al., 2018, Hornik et al., 1990). Such approximation does not on its own verify a returned dynamic policy or its numerical execution. Our certificates can be applied to neural or conventional continuations. The neural-specific construction lies in fresh own-future training and, in the coupled experiment, learned hidden locations. Its comparative value must be isolated against flexible conventional methods, not inferred from the certificate’s validity.

The general economic formulation admits recursive utility (Epstein and Zin, 1989, Duffie and Epstein, 1992). The compact-domain policy theorem uses monotonicity and cash invariance, while the original Gaussian entropic construction has separate moment and spectral conditions. These are different bridges, not interchangeable assumptions. The direct paired-cost identity below applies to conditional expec-

tation; it is not asserted for a nonlinear certainty equivalent. The confidence step uses the empirical Bernstein inequality of [Maurer and Pontil \(2009\)](#). Our additional ingredients are the policy-specific telescoping score, its certificate-derived support, and the numerical enclosure of the original continuous-law comparison.

2. THE CONTROLLED ECONOMY AND THE OPERATOR

2.1. Policy-specific continuation

The original continuous-time economy has an admissible control a and state

$$dS_t = b(t, S_t, a_t) dt + \sigma(t, S_t, a_t) dW_t. \quad (1)$$

The feasible correspondence, invariant or stopped domain, and exit and terminal rewards are economic primitives. For recursive evaluation, the continuation of a fixed policy is represented, on its specified horizon, by

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v.$$

The value is $V^a(t, s) = Y_t^a$ under the well-posedness and comparison hypotheses in the retained controlled-economy analysis. In particular, the recursive argument is the policy's own continuation. Replacing it by the optimal continuation would change the policy-evaluation problem.

The applications companion retains the full boundary specifications, infinite-horizon transversality conditions, utility-domain restrictions, temporal-self equations, and game conditions. A finite-domain residual calculation does not establish any of these by itself. In a game, certifying a specified deviation problem is not the same as establishing the equilibrium fixed point. In continuous time, the controlled generator contains state derivatives; freezing a critic during an actor update does not suppress legitimate derivatives of the represented state-dependent function.

For the constructive implementation and the main policy theorem, consider dates $t = 0, \dots, T$, compact state space X , and feasible correspondence $A_t(x)$. States evolve as

$$X_{t+1} = F_t(X_t, a_t, Z_{t+1}). \quad (2)$$

Let c_t be cost, g terminal cost, $\beta_t > 0$, $B_0 = 1$, and $B_t = \prod_{j < t} \beta_j$. We use a cost-minimization convention in the discrete implementation. It is equivalent to changing signs in an additive reward formulation, but does not replace the original recursive-utility assumptions by a sign convention.

Write

$$\begin{aligned} Q_t(v; x, a) &= c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z_{t+1}))), \\ \mathcal{T}_t v(x) &= \inf_{a \in A_t(x)} Q_t(v; x, a), \quad V_T^* = g. \end{aligned} \tag{3}$$

The conditional evaluations ρ_t are monotone and cash invariant on a domain containing every value and comparison function used below. Feasible measurable selectors exist; compactness, continuity of primitives, and continuity of the evaluated integrals are sufficient conditions in the experiments. The Bellman recursion is valid against all admissible adapted policies. For expectation, J^π is expected discounted cost. For a recursive evaluation it denotes the value generated by that evaluation's own recursion.

2.2. An executable NBO service

An implemented operator maps economic data and a current policy to

$$(\hat{f}, \hat{\pi}, \mathcal{C}, W) \quad \text{or} \quad (\text{unresolved}, W, \text{diagnostic}),$$

where $\mathcal{C}$ is an economic certificate and W records the work used. Its stages are target construction, fitting, feasible action selection, independent certification, and durable storage. Critic fitting freezes its future inputs; action selection freezes the resulting critic. In the fresh backward construction, date t uses the already fitted $\hat{f}_{t+1}$, not an externally solved optimal continuation. The terminal target is the primitive g .

A finite optimizer schedule is an algorithm, not a convergence theorem. A service that has not met its target at its cap records failure, with the complete cost of its attempts. The retained factor-training analysis provides a finite constructive theorem for a separate structured class. The nonlinear schedules below instead have an explicit, auditable capped-service estimand.

For a fixed catalogue $\mathcal{K}$ and target ε , define

$$H_m(k, \varepsilon) = \mathbf{1}\{\text{method } m \text{ stores a valid certificate at } \varepsilon \text{ within its cap on object } k\}. \quad (4)$$

The empirical object is the full array of H_m , costs to the first qualifying checkpoint, and capped costs for failures. Its arithmetic average is a catalogue fraction. A probability over random initializations would require a specified law and a sampling design for that different estimand. A post-review refinement of a failed object's verifier is a new diagnostic and does not change (4) for the original service.

3. CONTINUATION ACCURACY, TRANSPORT, AND POLICY LOSS

3.1. Centered errors and economic reuse

Consider a current objective $r_\theta(s, a) + S_\theta(T_\theta(s, a))$ to be maximized. Let $e = \hat{S}_0 - S_0$ be anchor approximation error and $\Delta_\theta = S_\theta - S_0$ the change in the actual future continuation. Suppose the implemented action is δ -optimal for $r_\theta + \hat{S}_0 \circ T_\theta$.

PROPOSITION 1—Centered decision and transport bounds: *The current decision loss satisfies*

$$L_\theta(s, \hat{a}) \leq \delta + \text{osc}_{A_\theta(s)}(e \circ T_\theta) + \text{osc}_{A_\theta(s)}(\Delta_\theta \circ T_\theta). \quad (5)$$

For a common finite action menu with positive weights p_j , set $R_p = \sum_j p_j(e_j - \sum_i p_i e_i)^2$ and $C_p = \max_{i \neq j} (p_i^{-1} + p_j^{-1})$. If $|\Delta_\theta| \leq b_\theta$ on the queried postdecision set, then $L_\theta \leq \delta + \sqrt{C_p R_p} + 2b_\theta$.

If common-innovation future transitions have Lipschitz constants L_k and perturbation bounds $\epsilon_{\theta,k}$, and future rewards have Lipschitz constants H_k and perturbations $\eta_{\theta,k}$ on an invariant domain, put $d_0 = 0$, $d_{k+1} = L_k d_k + \epsilon_{\theta,k}$. For nonnegative reward weights w_k ,

$$|S_\theta - S_0| \leq b_\theta := \sum_k w_k (\eta_{\theta,k} + H_k d_k). \quad (6)$$

The proof compares the true and fitted objective at the two selected actions and couples the future states. The weighted finite-menu bound is Cauchy–Schwarz applied to $e_i - e_j$. Details are in the technical supplement. An additive continuation error common to all actions is irrelevant to the decision. A finite menu, however,

does not certify a continuous action set merely because it includes both observed actions.

For recursive maps with sup-norm Lipschitz constants β_k on a common invariant utility domain, a one-step perturbation ν_k propagates as $b_k \leq \nu_k + \beta_k b_{k+1}$. This is the recursive counterpart of (6). Its use requires the common domain; it cannot cross a singular utility boundary by extending a local constant. These transport bounds retain the original refresh logic: use an anchor while its economic allowance covers approximation and future change, and recertify or recompute when that condition is exhausted.

3.2. A direct all-state policy theorem

For continuous fitted functions f_t and a feasible measurable policy π , suppose the verifier supplies, uniformly in x ,

$$\ell_t \leq f_t(x) - \mathcal{T}_t f_{t+1}(x) \leq u_t, \quad t < T, \quad (7)$$

$$Q_t(f_{t+1}; x, \pi_t(x)) - \mathcal{T}_t f_{t+1}(x) \leq \eta_t,$$

$$\ell_T \leq f_T(x) - g(x) \leq u_T.$$

These are enclosures for the actual coefficients and actual returned actions. Neither an optimizer's success flag nor a residual on sampled states replaces them.

THEOREM 2—Centered full-policy certificate: *Under the conditions of Section 2 and (7), every initial state satisfies*

$$0 \leq J_0^\pi(x) - V_0^*(x) \leq G(f, \pi) := \sum_{t < T} B_t(u_t - \ell_t + \eta_t) + B_T(u_T - \ell_T). \quad (8)$$

More specifically,

$$V_0^*(x) \geq f_0(x) - \sum_{t < T} B_t u_t - B_T u_T, \quad (9)$$

$$J_0^\pi(x) \leq f_0(x) + \sum_{t < T} B_t(\eta_t - \ell_t) - B_T \ell_T.$$

Replacing f_t by $f_t + d_t$, with date-specific constants d_t , leaves the bound unchanged after the residual endpoints are shifted by $d_t - \beta_t d_{t+1}$ and the terminal endpoints by d_T .

PROOF: Monotonicity and cash invariance imply $\mathcal{T}_t(v + c) = \mathcal{T}_t v + \beta_t c$ and the analogous identity for the fixed-policy map. Starting from $g \geq f_T - u_T$, backward induction gives the first inequality in (9). Starting from $g \leq f_T - \ell_T$ and using $Q_t(f_{t+1}; x, \pi_t(x)) \leq f_t(x) - \ell_t + \eta_t$ gives the second. Subtraction proves the upper bound; optimality proves the lower bound. The shifted residuals have unchanged widths and unchanged action gaps. *Q.E.D.*

The theorem certifies a full adapted policy, not a finite action menu or a grid of initial states. With expected cost its proof is a Bellman telescoping argument; for a monotone cash-invariant recursive evaluation it is an order argument. The numerical experiment is not needed for its validity. Its numerical usefulness depends on producing small, valid enclosures.

COROLLARY 3—Retained-policy recertification: *Fix the functions f_t , the deployed policy π , valid actor allowances η_t^{old} , and a terminal enclosure. Let a second verifier return valid residual enclosures $[\ell_t^{\text{new}}, u_t^{\text{new}}]$ for those same functions. Then*

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t^{\text{new}} - \ell_t^{\text{new}} + \eta_t^{\text{old}}) + B_T(u_T^{\text{old}} - \ell_T^{\text{old}}). \quad (10)$$

No retraining or change in deployed actions is required. If desired, intersecting two valid residual enclosures makes the new bound no larger than the old one.

The original actor allowance is valid on the entire state domain, so a finer critic verification mesh does not invalidate it. Conversely, replacing it by an allowance for a newly optimized fine-grid actor would certify a different policy. The implementation records the original actor hash and explicitly discards any fine-grid action proposals generated internally by the verifier. Those proposals' work and transient storage are still charged.

The corollary does not promise that arbitrary precision is attainable by mesh refinement. Nodal residual variation, the retained actor allowance, and the terminal enclosure can leave a positive floor. Nor is the corollary a claim that additional verification improves welfare. It distinguishes the accuracy of a fixed economic decision rule from the sharpness of our knowledge of that accuracy.

3.3. Construction beyond post-training verification

The original trainable-factor construction remains a constructive part of NBO. For a quadratic own-policy continuation $x'P_tx + c_t$, a square-activation network $\|W_tx\|^2/d + \hat{c}_t$ trains every entry of W_t . Its population derivative loss and gradient are

$$\mathcal{L}_t(W; P) = \frac{1}{4} \|W'W - dP\|_F^2, \quad \nabla_W \mathcal{L}_t = W(W'W - dP).$$

If $\sigma_{\min}(W) \geq s > 0$, then $\|W'W/d - P\|_F \leq \|\nabla_W \mathcal{L}_t\|_F / (ds)$. Together with the full-vector actor defect, this gives a policy-error account for a jointly trained hidden factor. The own-policy coefficient is re-evaluated for the returned policy, not silently borrowed from an earlier training policy.

The primitive-budget analysis in the retained article chooses a feasibility witness, coefficient envelopes, rank margins, and update budgets before training. In the Gaussian entropic extension it also verifies positive exponential-moment margins. Under its stated spectral hypotheses, a scaled orthogonal initialization and bounded gradient steps give the finite training construction proved there. The finite matrix and Gaussian formulas are not a nonlinear dimension-scaling experiment. They also do not imply that Adam or L-BFGS has a deterministic residual cap for a general squared-ReLU network. We preserve both results and keep the two logical statements separate.

4. FROM ECONOMIC PRIMITIVES TO NONLINEAR CERTIFICATES

4.1. Backward fitting and selected policies

For each method, initialize a fresh terminal fit from g . At date t , construct targets from $\mathcal{T}_t f_{t+1}$ using that method's own already fitted future. Fit the current continuation under the frozen schedule. Then propose feasible actions, enclose the Bellman infimum and the selected-action gap, and validate residual coverage. Store a checkpoint before reporting its first target crossing. Continue through the declared ladder only as specified by the original study. The record includes all attempted resolutions, unsuccessful target queries, target-generation work, and storage operations.

The scalar network has 63 ReLU features. Its training uses 513 states, a 257-point target action construction, 400 Adam steps at learning rate 5×10^{-5} , and a final output fit. The scalar verifier uses its own finer action and state ladders. The conventional spline is reconstructed from the same primitives with its own ladder and targets. A conventional target solver is not used to label a neural continuation.

The coupled neural continuation is

$$f_t(x) = a_t + b'_t x + \sum_{j=1}^{48} c_{tj} (w'_{tj} x + v_{tj})^2_+, \quad c_{tj} \geq 0. \quad (11)$$

Hidden locations and weights are trained. Nonnegative coefficient initialization and a finite L-BFGS schedule of 300 iterations act on fresh own-future targets. The ridge comparator instead fixes a 96-feature squared-hinge dictionary and fits nonnegative coefficients to its own targets; zero coefficients may be pruned. Its dictionary seed is fixed by date, so repeated training-seed labels do not create independent random ridge representations. A quadratic projection is retained as a restricted comparator, not treated as the strongest conventional alternative.

4.2. All-state and continuous-action verification

At each state node, outward arithmetic encloses the actual network value and the action objective. In the coupled economy, convexity in actions and a positive strong-convexity constant yield a continuous-simplex primal–dual gap. A projected optimizer merely proposes the action; its stopping flag is not the certificate. State interpolation then extends the signed residual bounds off-grid. The nearest-node deployed actor receives a separate, uniform allowance for using its node action at the actual state.

Uniform innovations are integrated through positive-part moments, rather than replaced by a sampled innovation law. The elementary identity for a uniform variable U on $[-h, h]$ is

$$\mathbb{E}(y + U)^2_+ = \frac{(y + h)^3_+ - (y - h)^3_+}{6h}, \quad h > 0.$$

Successive integration gives the corresponding formula for independent coordinates and affine neuron arguments; zero coefficients are handled by the lower-dimensional formula. Interval evaluation encloses cancellation and rounding. The supplement specifies the mesh and action bounds used in the inherited verifier.

The residual account distinguishes a nodal enclosure from state coverage, actor error, and terminal error. A nodal enclosure contains the effects of representation, fitting, Bellman optimization, and arithmetic; it is not an identified decomposition into a pure approximation error and a pure optimizer error. In particular, subtracting a training loss from the residual does not estimate either component. The diagnostic reports every separately available allowance without counting nested subcomponents twice.

4.3. Work, dimension, and execution

For tensor state, action, and innovation covers of sizes K_x, K_a, m , a direct verifier pays work of order

$$\sum_{t < T} K_x K_a \{m(C_F + P_{t+1}) + C_\rho\} + \sum_{t \leq T} K_x P_t + W_{\text{storage}}, \quad (12)$$

where P_t counts network evaluation work. Actor storage is of order $TK_x d_a$. Analytic integration replaces m by the work of the moment formula, not by zero. With tensor covers, K_x scales as $h_x^{-d_x}$; analogous factors arise for action and innovation discretizations. A dimension-explicit certificate is therefore not a dimension-free algorithm.

If a Lipschitz innovation approximation contributes at most ζ_t to Q_t , widening residual endpoints by ζ_t and the actor allowance by $2\zeta_t$ adds $4 \sum_t B_t \zeta_t$ to (8). This is a discretization account, not permission to replace a continuous law without charging its error. Independent policy queries may have their own scenario-tree or simulation cost and are recorded separately.

The complete service includes own-future targets, fitting, unsuccessful refinement, action extraction, verification, applicable policy queries, serialization, and durable checkpoint storage. Initialization and outer-process time are retained separately. New local diagnostic clocks are not added to old GitHub-hosted clocks to

manufacture an end-to-end estimate. All reported arithmetic certificates concern stored binary coefficients with outward enclosures. The inherited execution account treats rounded state acquisition and transition/cost evaluation separately from ideal policy loss.

5. DIRECT COMPARISONS OF ECONOMIC POLICY COSTS

A smaller upper bound on $J^\pi - V^*$ need not mean a smaller J^π . We therefore compare the neural and ridge policies directly at common states, using common innovations and the expectation version of (3).

For method m , let X^m follow its own fixed policy and define

$$Z_m = f_0^m(x_0) + \sum_{t < T} B_t \{Q_t(f_{t+1}^m; X_t^m, \pi_t^m(X_t^m)) - f_t^m(X_t^m)\} + B_T \{g(X_T^m) - f_T^m(X_T^m)\}. \quad (13)$$

This is a control-variate score for actual policy cost, not the difference of two regret bounds.

THEOREM 4—Paired economic score and support: *For fixed policies and integrable continuations under conditional expectation, $\mathbb{E}Z_m = J_0^{\pi^m}(x_0)$. If method m satisfies (7), then every trajectory satisfies*

$$L_m \leq Z_m \leq U_m, \quad (14)$$

$$L_m = f_0^m(x_0) - \sum_{t < T} B_t u_t^m - B_T u_T^m,$$

$$U_m = f_0^m(x_0) + \sum_{t < T} B_t (\eta_t^m - \ell_t^m) - B_T \ell_T^m.$$

Consequently $Z_N - Z_R$ has expectation $J^{\pi^N} - J^{\pi^R}$ and support $[L_N - U_R, U_N - L_R]$, whose width is $G_N + G_R$. These statements hold under any coupling of the two methods that preserves each policy's innovation law.

PROOF: Conditional expectation gives $\mathbb{E}Q_t(f_{t+1}^m; X_t^m, \pi_t^m) = \mathbb{E}c_t + \beta_t \mathbb{E}f_{t+1}^m(X_{t+1}^m)$. Summing telescopes all continuation terms. The selected-action gap is between zero and η_t^m , so each centered summand in (13) is between

$-u_t^m$ and $\eta_t^m - \ell_t^m$. The terminal summand is between $-u_T^m$ and $-\ell_T^m$. Summing
proves the support statement and its width. Q.E.D.

To evaluate the original continuous law, simulate uniform innovation bins and
enclose all within-bin innovations. Interval propagation includes every actor that
can be selected when a state enclosure crosses a nearest-node boundary. Suppose the
resulting endpoint pair satisfies $A_i \leq Z_{N,i} - Z_{R,i} \leq D_i$ and lies in a known support
interval of width B . For independent draws of bins, the lower and upper endpoints
are bounded independent observations. Let $\bar{A}, \bar{D}$ and s_A^2, s_D^2 be their sample means
and variances. For q predeclared comparisons, set

$$r(s^2) = \sqrt{\frac{2s^2 \log(4q/\alpha)}{n}} + \frac{7B \log(4q/\alpha)}{3(n-1)}. \quad (15)$$

Applying the one-sided empirical Bernstein inequality to the lower and upper
endpoints separately, then taking a union bound, gives simultaneous coverage at
least $1 - \alpha$ for

$$[\bar{A} - r(s_A^2), \bar{D} + r(s_D^2)] \quad (16)$$

over all q comparisons. Mean, variance, logarithm, and square-root calculations
are outward enclosed in the implementation; intersecting with the deterministic
support preserves coverage. The proof and numerical details are in the supplement.

The coverage statement is conditional on frozen training outputs and an in-
dependent simulation model. Fixed PCG64 streams make this implementation
reproducible; a fixed stream is not itself a proof of probabilistic independence. No
policy is selected using these simulations. Intervals crossing zero are reported as
unresolved, not as equivalence or as evidence of a superior neural policy.

6. INVESTMENT ECONOMIES AND THE COMPLETE EVIDENCE

6.1. Design and provenance

The original science design has twelve scalar neural services, twelve scalar
spline services, six coupled neural services, six coupled ridge services, six coupled
quadratic services, and 36 services for each of three precision rules. The successful

source-bound execution contains all 150 records. The original first run, with four nonfinite-network exceptions, is retained. Its numerical inverse-softplus initialization was subsequently replaced by an algebraically equivalent stable expression; economic primitives, seeds, targets, budgets, and comparators were unchanged. We do not treat the failed first run as nonexistent or combine its clocks with the successful run.

The R44 protocol was committed before the additional diagnostics were executed. It is a post-review protocol, not an original preregistration. It selects the seven unresolved tight-target neural objects by their already observed outcomes and freezes their weights and policies. The paired comparison uses all six coupled pairs and all four originally declared states. A source manifest binds the original inputs, the diagnostic code, outcomes, and publication sources. Every original record and all five original science-source hashes were replayed.

6.2. *Scalar and coupled primitives*

The scalar experiment retains the original bounded investment state, price and risk cells, three declared training seeds, continuous feasible action interval, and the independent own-policy evaluator. Prices are one and four. Both methods construct their own continuations from the same primitive problem. The retained evidence and code specify every scalar primitive; the full economic derivation remains in the original investment section and applications.

In the coupled economy, $x \in [0, 1]^2$, $T = 4$, $\beta = 15/16$, and $A = \{a \geq 0 : a_1 + a_2 \leq 1/8\}$. With independent $z_j \sim U[-1/32, 1/32]$,

$$F_j(x, a, z) = \frac{1}{8} + \frac{1}{2}x_j + \frac{1}{8}x_{3-j} + \frac{1}{16}x_j(1 - x_{3-j}) + a_j + z_j.$$

This transition maps the stated state-action-innovation domain into the state domain. Define

$$s(x) = \sum_{j=1}^2 (x_j - 11/16)^2 + 2(1/2 - x_1 - x_2)_+^2 + \frac{1}{4}(x_1 - x_2)^4.$$

The stage cost is $s(x) + p \sum_j a_j^2 + \frac{1}{2}a_1a_2 + 4 \sum_j a_j^4$; terminal cost doubles the squared-target term in s and retains its remaining terms. Prices $p \in \{1, 4\}$ penalize costly

investment. The shared simplex represents a common resource constraint, the transition couples the two capital stocks, and the state cost penalizes aggregate shortage and imbalance. This is a stylized controlled economy, not an empirically calibrated welfare exercise.

For both coupled methods the final all-state verification ladder is $N = 16, 32, 64, 128$. The action objective has strong-convexity lower bound $2p - 1/2 > 0$. Uniform positive-part moments integrate continuous uncertainty analytically. The finest original actor has $4(129)^2 \cdot 2 = 133,128$ stored scalar actions. This actor cost is distinct from continuation-network storage.

6.3. Original attainment and comparative outcomes

TABLE I

ORIGINAL CAPPED-SERVICE TARGET ATTAINMENT

Method	All-state target	Certified
Scalar neural	0.06	12/12
Scalar neural	0.04	6/12
Scalar spline	0.06	12/12
Scalar spline	0.04	12/12
Coupled neural	0.25	6/6
Coupled neural	0.10	5/6
Coupled ridge	0.25	6/6
Coupled ridge	0.10	6/6
Coupled quadratic	0.25	0/6
Coupled quadratic	0.10	0/6

Table I reports the original capped-service outcomes. The original scalar neural bound is larger than its matched spline bound in every one of the twelve pairs. At the first common 0.06 crossing, 29 of 48 signed neural-minus-spline intervals have positive lower endpoints, none has a negative upper endpoint, and 19 contain zero. At 0.04, the six common pairs give 17 positive intervals, none negative, and seven overlaps. All lie within the prespecified descriptive margin ± 0.005 . This establishes

numerical closeness at those states but is not an equivalence design; where signs are identified, they favor the spline.

The coupled ridge reaches both targets in all six services. It has the smaller final gap in four pairs and the neural method in two. At 0.25 the ridge reaches the target faster in all six original observations. At 0.10 it is faster in four of the five pairs in which both pass, while the neural method is faster in one and fails in the remaining pair. These singleton nonlinear timing observations are descriptive. The ridge continuation has storage comparable to the neural continuation, so its strength cannot be dismissed as a comparison with an artificially restricted quadratic.

6.4. Diagnosing tight-target failures without changing policies

TABLE II

ADDITIONAL VERIFICATION OF UNCHANGED NEURAL POLICIES

Service	Original bound	Refined bound	Target	Attained
041	0.04092128	0.03068472	0.04	Yes
089	0.04118465	0.03090917	0.04	Yes
120	0.04112077	0.03085353	0.04	Yes
124	0.04093160	0.03070697	0.04	Yes
128	0.15223570	0.09775755	0.10	Yes
145	0.04136567	0.03108538	0.04	Yes
147	0.04101560	0.03077643	0.04	Yes

For the six unresolved scalar policies, only the residual state grid is refined from $N = 2048$ to $N = 4096$, retaining the original action verification resolution $A = 1024$. For the coupled policy, residual verification uses $N = 256$ instead of 128. The original actor allowances and terminal enclosures remain in (10); all original action arrays and networks are hash-checked unchanged. Fine-grid actor proposals generated internally are discarded, not deployed.

Table II gives all seven results. The retained scalar policy bounds become approximately 0.03068–0.03109, below 0.04. The retained coupled policy bound becomes 0.09775756, below 0.10. There are zero training updates and zero newly stored

deployed-action scalars. The coupled diagnostic still pays for a fine verification mesh with 66,049 states, including transient fine-grid proposals; retaining the original actor does not make verification free.

The original coupled decomposition is informative. Its discounted state-cover contribution to nonterminal residual widths is about 0.07268224, its actor allowance is 0.03287249, its remaining nodal residual enclosure is 0.04574738, and its terminal contribution is 0.00093360. These sum to the original bound up to displayed rounding. The extra mesh quarters the principal second-order coverage term while retaining the actor and terminal terms. The resulting crossing does not establish that representation and optimization errors are separately small; it establishes the required full-policy bound for the unchanged policy.

The diagnostic answers a specific method question. The original capped service was less reliable at its tight target, but the saved neural objects were not necessarily inadequate policies. All seven selected policies admit the tighter certificates when verification is extended under a fixed rule. This finding does not alter the original attainment table or justify a comparison between a selectively refined neural certificate and an unrefined comparator as evidence of a work advantage.

6.5. *Direct coupled economic differences*

TABLE III

DIRECT NEURAL-MINUS-RIDGE COST COMPARISONS, SIMULTANEOUS 99 PERCENT INTERVALS

Price	Comparisons	Positive	Negative	Unresolved
1	12	0	0	12
4	12	0	0	12

For each of the six original neural/ridge pairs, select each method's first 0.25 crossing. Compare at $(1/8, 1/8)$, $(1/4, 1/2)$, $(1/2, 1/4)$, and $(3/4, 3/4)$. Each comparison uses 65,536 common innovation-bin trajectories, 40-bit uniform bins, and the own-policy score in (13). The total family error is 0.01 across 48 one-sided bounds. The complete 24 intervals, endpoint moments, seeds, ambiguity counts, and support bounds appear in the supplement and machine-readable records.

All 24 direct confidence intervals contain zero. Thus the coupled evidence now addresses the actual neural-minus-ridge policy-cost difference, but does not rank the methods. This is stronger information than subtracting regret bounds and different information from a smaller all-state certificate. The simultaneous interpretation pertains to the declared states and fixed policies under the simulation model, not to every state in the compact domain. The all-state policy guarantees remain the separate deterministic statements in Table I and Theorem 2.

6.6. Precision and complete recorded work

TABLE IV
ORIGINAL REPEATED PRECISION CATALOGUE

Method	Certified	Accepted32	Rejected32	Accepted64
Fixed32	6/36	147	30	0
Fixed64	36/36	0	0	474
Adaptive	36/36	348	126	126

The precision experiment repeats twelve dimension/conditioning/target objects three times. Fixed binary64 and adaptive each certify all 36 services; fixed binary32 certifies six. Fixed64 and adaptive each make 474 accepted updates. Adaptive makes 348 accepted binary32 updates, 126 rejected binary32 proposals, and 126 accepted binary64 updates. Every rejection is charged.

Adaptive is slower in 26 of the 36 matched recorded services and faster in ten; aggregate recorded time is about 0.98% larger. The controller’s activation therefore demonstrates its numerical mechanism, not a complete-work saving in this implementation. This result does not invalidate the conditional numerical acceptance theorem. It prevents the theorem from being mistaken for an empirical efficiency conclusion.

6.7. What the evidence identifies

The evidence identifies a finite from-primitives nonlinear construction, all-state certificates for its returned policies, a useful retained-policy verification refinement,

and direct economic comparisons under the declared state and innovation design. It also identifies the continued strength of conventional comparators. It does not identify a general neural speed advantage, a nonlinear high-dimensional frontier, or a population probability of optimizer success.

An adaptive sparse-grid or adaptive-partition comparator has not been executed in this revision. Neither has a nonlinear dimension/horizon grid or repeated isolated timing of every scalar and coupled service. The retained dimension-explicit analysis exposes the tensor factors those studies would have to confront, rather than treating a two-state experiment as their substitute. The general economic framework and conditional constructive results remain in force; the missing experiments concern comparative evidence for broader implementation claims.

7. ECONOMIC APPLICATIONS AND CONCLUSION

The original applications concern policy-dependent recursive utility, endogenous preference specifications, temporal selves, controlled diffusion problems, and dynamic games. Their complete economic statements and proofs are retained without alteration. The centered and transport bounds explain how an NBO can support economic counterfactuals when changes in future primitives are controlled. The policy theorem explains what must be verified for the actual policy. Neither statement supplies an equilibrium or transversality condition absent from the application.

The nonlinear investment evidence adds an executable link from primitives to learned continuations and directly certified policies. Its retained-policy diagnostic demonstrates that a failed tight certificate can reflect the resolution of verification rather than the quality of the deployed policy. Its direct coupled comparison answers a different question—the relative cost of two policies—without confusing separate regret bounds with a method contrast.

Neural Bellman Operators thus organize construction, certification, economic reuse, and resource accounting around the policy actually returned. Their scientific assessment requires all of these objects. Flexible conventional methods remain important competitors; numerical activation is not a work saving, and a fixed catalogue is not a general reliability theorem. Preserving those distinctions makes the original economic program testable while retaining its full theory and applications.

APPENDIX A: READING AND PRESERVATION MAP

The present main article is the authoritative R44 exposition. The accompanying technical supplement gives complete proofs and the full new comparison table. The response addresses every B1–B10 and M1–M10 comment. The unchanged R41 article preserves all original sections, theorem statements, proofs, and historical evidence; its 45-page supplement and 48-page economic-applications companion remain linked in the release. A machine-readable preservation audit records byte hashes and all original LaTeX labels. Previous revisions and review branches are untouched.

The unchanged files are retained content, not an alternative authoritative description of the new experiment. Where an older empirical description and the present report differ, the source-bound R42 catalogue and the explicitly separated R44 diagnostics control the empirical statement. This precedence avoids deleting history while providing one current manuscript.

APPENDIX: REFERENCES

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